Module: Java Setup and First Programs

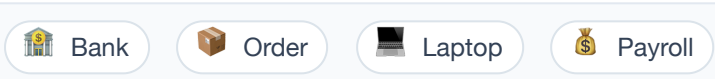
Phase: Core Java
Previous module: Computer and Programming Foundations
Next module: Variables, Datatypes, Operators, I/O

Module Outcome:

- Install JDK 17, compile with `javac`, run with `java` — no mystery IDE button
- Write `Hello` as a class with `main` and explain bytecode + JVM in one line
- Know why the same `.class` runs on Windows, macOS, and the lab PC

Why Java?

Java is a programming language first released in 1995. It is **compiled** to **bytecode**, then run by the **Java Virtual Machine (JVM)** — so the same `.class` files can run on Windows, macOS, and Linux.



Where used?

- **SecureBank** / UPI — welcome banners, balance printouts in training labs
- College apps — student portal banners, marks entry CLIs (syntax first — no named product required)
- **TechMart** — order labels and first checkout demos
- **AcmeHR** — employee welcome scripts and HR training labs
- **MedCare** — appointment “hello” scripts before real CRUD

Why companies in India hire Java developers:

- **Backend APIs:** large product and services teams still list Core Java + REST
- **Android history:** many mobiles run on the Java language family (this course is **server/full-stack**, not Android)
- **Enterprise hiring:** banks, product companies, and service firms interview Core Java first
- **Full stack:** Java API + React frontend is a common job stack

This course uses **JDK 17 LTS** (Long-Term Support). **JDK 21** is also LTS — mention it on JDs; class default is 17.

JDK vs JRE: the **JDK** (Java Development Kit) includes the compiler `javac`. The **JRE** (runtime) can run programs but cannot compile. Install the **JDK**.

Installing JDK 17

- macOS: `brew install openjdk@17` (then follow the PATH hint Homebrew prints)
- Windows: Eclipse Temurin 17 MSI — tick **Add to PATH**
- Verify:

```
java -version
javac -version
```

Expected Output: (wording varies by vendor)

```
openjdk version "17.x.x" ...
javac 17.x.x
```

If `java` works but `javac` is missing, you installed a JRE, not a JDK.

IDE for Java

1. **VS Code:** Extension Pack for Java
2. **IntelliJ Community:** File → Project Structure → SDK 17

Open folder `ksr_java_fullstack`. Create `projects/HelloJava.java`.

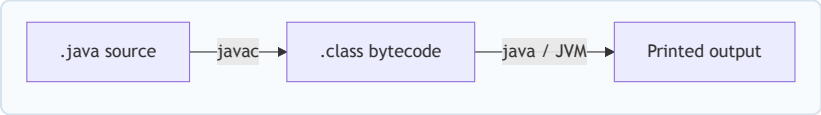
Compile then run

Java is **not** "save and interpret one line" like some languages. You:

1. Write `HelloJava.java`
2. Compile: `javac HelloJava.java` → creates `HelloJava.class`
3. Run: `java HelloJava` (no `.class` in the command)

```
HelloJava.java  --javac-->  HelloJava.class  --java-->  output on screen
     source                   bytecode                    JVM
```

Visual — compile vs run:



Key takeaways:
- You edit `.java`; the machine runs `.class`
- `javac` must succeed before `java`
- Class name and file name match for `public class`

Keywords and identifiers

Keywords are reserved words Java uses for grammar — you cannot name a variable `class` or `int`.

Identifiers are names you choose: classes, variables, methods.

Rules:

- Start with a letter or underscore `_` (not a digit)
- Rest can be letters, digits, underscore
- **Case-sensitive:** `name` and `Name` are different
- No spaces: `studentName` not `student name`
- Convention: **PascalCase** for class names (`HelloJava`); **camelCase** for variables (`studentName`)

Important Concepts and Definitions

Term	Definition
JDK 17	Development kit this course pins (compiler + runtime)
JVM	Virtual machine that runs bytecode
Bytecode	`.class` output of `javac`
`javac`	Compiler command
`java`	Launcher that starts the JVM
`main`	Entry point the JVM looks for
Keyword	Reserved word (`class`, `public`, `static`, `void`)
Identifier	Name you choose
PascalCase	`HelloJava`, `StudentMarks` — classes
camelCase	`rollNumber`, `isPass` — variables / methods
Comment	Ignored by the compiler (`//` or `/* */`)
PATH / JAVA_HOME	Where the OS finds `java` and `javac`
jshell	Optional REPL shipped with the JDK

Step-by-Step Notes (With Annotated Code)

Step 1: Environment checklist

Create `projects/EnvCheck.java`:

```
public class EnvCheck {
    public static void main(String[] args) {
        System.out.println("Java version check from your terminal, not from this print.");
        System.out.println("Setup OK — ready for Core Java!");
        System.out.println("Priya — KSR Java Full Stack");
    }
}
```

Run:

```
cd ~/ksr_java_fullstack/projects
javac EnvCheck.java
java EnvCheck
```

Note:
- Wrong: Running `java EnvCheck.java` as the only habit and never learning `javac` (interviewers still ask compile vs run).

- Prefer: `javac` then `java ClassName` on the command line at least once a day this week.
- Wrong: `JAVA_HOME` pointing at Java 8 while class requires 17.
- Prefer: `java -version` shows **17**.

Print a ready line that includes Priya's name.

```
public class ReadyLine {
    public static void main(String[] args) {
        System.out.println("Ready for KSR Java — Priya");
    }
}
```

Expected Output:

```
Ready for KSR Java — Priya
```

Print a ready line that includes Priya's name and the city Hyderabad.

```
public class ReadyCity {
    public static void main(String[] args) {
        System.out.println("Ready for KSR Java — Priya (Hyderabad)");
    }
}
```

Expected Output:

```
Ready for KSR Java — Priya (Hyderabad)
```

Step 2: jshell — try before you save (optional)

```
jshell
```

```
jshell> System.out.println("Hello from jshell")
Hello from jshell
jshell> 2 + 3
$1 ==> 5
jshell> /exit
```

jshell is great for one expression. **Assignments are `.java` files.** If `jshell` is missing, skip it — scripts still count.

Print a greeting for Ravi and the sum of two TechMart canteen prices (₹450 and ₹120).

```
public class CanteenSum {
    public static void main(String[] args) {
        System.out.println("Hello, Ravi");
        System.out.println(450 + 120);
    }
}
```

Expected Output:

```
Hello, Ravi
570
```

Print the learner name, the batch name, and a line that confirms Java 17 is OK.

```
public class BatchOk {
    public static void main(String[] args) {
        System.out.println("Priya");
        System.out.println("Full Stack Java Batch");
        System.out.println("Java 17 OK");
    }
}
```

Step 3: First script — hello world

`projects/HelloJava.java`:

```
// HelloJava.java
// Author: Your Name
// Purpose: First Java program for KSR Full Stack course

public class HelloJava {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
        System.out.println("Welcome to Java Full Stack training.");
    }
}
```

```
javac HelloJava.java
java HelloJava
```

Annotation:

- `public class HelloJava` — class name **must match** the file name
- `main` — JVM starts here
- `System.out.println(...)` — print a line
- `//` starts a **comment**

Note:
- Wrong: `public class hellojava` in file `HelloJava.java` → compiler error.
- Prefer: File `HelloJava.java` and `public class HelloJava`.
- Wrong: `java HelloJava.class` → error. Use `java HelloJava`.

Print Priya's name and one-line career goal.

```
public class MyGoal {
    public static void main(String[] args) {
        System.out.println("Priya");
        System.out.println("Become a full-stack Java developer");
```

```
    }
}
```

Expected Output:

```
Priya
Become a full-stack Java developer
```

Print the same name and career goal, and add author and date comments above the code.

```
// Author: Priya
// Date: 23-08-2026
public class MyGoalCommented {
    public static void main(String[] args) {
        System.out.println("Priya");
        System.out.println("Become a full-stack Java developer");
    }
}
```

Step 4: Multi-line output — training centre welcome

```
public class WelcomeStudent {
    public static void main(String[] args) {
        System.out.println("========================================");
        System.out.println("   KSR STUDENT PORTAL");
        System.out.println("========================================");
        System.out.println("Login page loading...");
        System.out.println("Please use registered email.");
        System.out.println("========================================");
    }
}
```

Expected Output:

```
========================================
   KSR STUDENT PORTAL
========================================
Login page loading...
Please use registered email.
========================================
```

Print a SecureBank ATM welcome banner using repeated `=` lines around the title and welcome text.

```
public class AtmBanner {
    public static void main(String[] args) {
        System.out.println("========================");
        System.out.println("    SECUREBANK ATM");
        System.out.println("========================");
        System.out.println("  Welcome, customer");
        System.out.println("========================");
    }
}
```

Expected Output:

```
========================
    SECUREBANK ATM
========================
  Welcome, customer
========================
```

Print a student marks portal banner: title, a blank line, then a prompt to enter a roll number.

```
public class MarksBanner {
    public static void main(String[] args) {
        System.out.println("STUDENT MARKS PORTAL");
        System.out.println();
        System.out.println("Enter roll to continue");
    }
}
```

Step 5: Keywords — what you cannot use as names

```
public class KeywordDemo {
    public static void main(String[] args) {
        // INVALID if uncommented — keywords:
        // int class = 1;
        // int public = 2;

        int classCount = 1;
        int publicFlag = 2;
        System.out.println(classCount);
        System.out.println(publicFlag);
    }
}
```

Expected Output:

```
1
2
```

Note:
- Wrong: `int class = 10;` → compile error.
- Prefer: `classCount`, `isPublic`.

Store `if`, `class`, and `for` as legal identifier names (`ifFlag`, `className`, `forCount`) and print each value.

```
public class ValidNames {
    public static void main(String[] args) {
        int ifFlag = 1;
        String className = "A";
        int forCount = 5;
        System.out.println(ifFlag);
        System.out.println(className);
        System.out.println(forCount);
```

```
    }
}
```

Print a few Java keywords as text (not as variable names), then a line that says they are reserved.

```
public class KeywordList {
    public static void main(String[] args) {
        System.out.println("class public static void int");
        System.out.println("These words are reserved — do not use them as names.");
    }
}
```

Step 6: Identifier naming — employee portal style

```
public class NamingStyle {
    public static void main(String[] args) {
        String fullName = "Ananya Sharma";
        String departmentCode = "HR";
        boolean isActive = true;

        System.out.println(fullName);
        System.out.println(departmentCode);
        System.out.println(isActive);
    }
}
```

Expected Output:

```
Ananya Sharma
HR
true
```

Note:
- Wrong: `String FullName` for a variable — works, but looks like a class.
- Prefer: `fullName` for variables; `FullName` only for types (later, OOP).

Common Error: Space in a name (`int employee id = 101`) → compile error. Use `employeeId` .

Store a student name, roll number, and pass flag with clear `camelCase` names, then print each value.

```
public class StudentIds {
    public static void main(String[] args) {
        String studentName = "Priya";
        String rollNumber = "2026001";
        boolean isPass = true;
        System.out.println(studentName);
        System.out.println(rollNumber);
        System.out.println(isPass);
    }
}
```

Expected Output:

```
Priya
2026001
true
```

Store food expense as `foodExpense` and category as `categoryName`, then print `food: ₹250`.

```
public class ExpenseNames {
    public static void main(String[] args) {
        int foodExpense = 250;
        String categoryName = "food";
        System.out.println(categoryName + ": ₹" + foodExpense);
    }
}
```

Expected Output:

```
food: ₹250
```

Step 7: `main` signature — do not invent it

The JVM looks for **exactly**:

```
public static void main(String[] args)
```

`String[] args` is a list of command-line words. You can ignore it until arrays week.

Remember: Misspell `Main` or drop `static` and the program will not start the way you expect.

Print a one-line TechMart canteen bill title.

```
public class CanteenBill {
    public static void main(String[] args) {
        System.out.println("TechMart Canteen Bill");
    }
}
```

Print the same TechMart canteen title, then a second line that says to pay at the counter.

```
public class CanteenBillTwo {
    public static void main(String[] args) {
        System.out.println("TechMart Canteen Bill");
        System.out.println("Pay at counter");
    }
}
```

Step 8: Workflow (recommended habit)

```
[Think] → [Type in HelloJava.java] → [javac] → [java] → [Git commit later]
```

Note:
- Wrong: Running from the wrong folder → `error: file not found`.
- Prefer: `cd` to the folder that contains the `.java` file.

Compile `HelloJava.java` and list the generated `HelloJava.class` to confirm the class file exists.

```
javac HelloJava.java
ls HelloJava.class
```

Run the compiled `HelloJava` class after `javac` succeeds.

```
java HelloJava
```

Practice Exercises

Basic

1. Install JDK 17 and paste terminal output of `java –version` and `javac –version` into `notes/setup_proof.txt`.

Model answer: After install, run:

`bash java –version javac –version`

Both should show **17**. Paste both blocks into `notes/setup_proof.txt`.

2. Write `HelloJava.java` that prints your name and city on two lines.

Model answer:

```
public class HelloJava {
    public static void main(String[] args) {
        System.out.println("Priya Sharma");
        System.out.println("Hyderabad");
    }
}
```

Then `javac HelloJava.java` and `java HelloJava`.

3. In jshell (or a tiny class), print `10 * 5` and `100 / 4`.

Model answer: `10 * 5` is `50`. `100 / 4` is `25` because both sides are `int` (integer division). Float division is next module.

4. List 8 Java keywords from this module.

Model answer: `class`, `public`, `static`, `void`, `int`, `boolean`, `true`, `false` (also `if`, `for` later — do not use them as names).

Intermediate

1. Write `TrainingBanner.java` that prints a 5-line banner for "Employee Portal Training".

Model answer:

```
public class TrainingBanner {
    public static void main(String[] args) {
        System.out.println("============================");
        System.out.println("  Employee Portal Training");
        System.out.println("  KSR Full Stack Java");
        System.out.println("  Welcome, learners!");
        System.out.println("============================");
    }
}
```

2. Fix this code (find 3 errors):

`java public class Demo { public static void Main(String[] args) { System.out.Println("Hello"); int class = 1; } }`

Model answer: (1) `Main` must be `main`. (2) `Println` must be `println`. (3) `class` is a keyword — rename to `classCount`.

3. Create identifiers for: student roll number, monthly salary, is_logged_in flag.

Model answer: `rollNumber` (or `studentRollNumber`), `monthlySalary`, `isLoggedIn`. Classes stay PascalCase.

4. Compile from the wrong folder once, then fix with `cd`. Note what error you saw.

Model answer: `javac: file not found: HelloJava.java` (wording varies). `cd` into `projects/` (or pass the full path) and compile again.

Interview-Oriented

1. Is Java compiled or interpreted? Explain in 2–3 sentences.

Model answer: Java is **compiled** to bytecode (`javac`), then the **JVM interprets/JIT-compiles** that bytecode. Short interview line: “Compiled to bytecode, runs on the JVM.”

2. Why must `public class` name match the file name?

Model answer: The compiler and tooling look up the class by file. A mismatch is a compile error for public types. Convention keeps one public class per file.

3. JDK vs JRE?

Model answer: JDK = compiler + tools (`javac`) + runtime. **JRE** = runtime only. Developers install the JDK.

4. What happens if you write `void main()` without `static`?

Model answer: The JVM looks for `public static void main(String[])`. Without `static` it will not use that method as the entry point (error at launch).

Mini-Project (Assignment): Environment and Greeting Pack

Lab kit: 15_Lab_Assets/02_hello_javac/

What to build

1. `projects/EnvCheck.java` — prints a ready message
2. `projects/HelloJava.java` — hello world
3. `projects/MyIntro.java` — name, career goal, one reason you chose Java full stack
4. `notes/setup_proof.txt` — `java -version` and `javac -version`

5. Screenshot: terminal after `javac` + `java MyIntro`

Stretch

- Create a GitHub account and push `HelloJava.java` (empty repo is enough this week).
- Add `WelcomeBatch.java` that prints a 5-line KSR banner.

Done when

- [ ] `java -version` and `javac -version` show 17
- [ ] `EnvCheck`, `HelloJava`, and `MyIntro` compile and run
- [ ] Lab kit ticket completed
- [ ] No `.class` files needed in Git later — for now, running locally is enough

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Interview Q&A

Shared with students after concepts and the mini-project. Use for review and interview prep — questions and model answers are included.

Q1. What is Java? What is it used for?

A: A high-level language that compiles to bytecode and runs on the JVM. Used for backend APIs, enterprise systems, Android-family apps, and full-stack servers.

Q2. Difference between `javac` and `java`?

A: `javac` compiles `.java` → `.class`. `java` starts the JVM and runs the class.

Q3. What are Java keywords? Example of invalid use?

A: Reserved words like `class`, `public`, `static`. `int class = 5;` will not compile.

Q4. How do you run a Java file from the command line?

A: `javac HelloJava.java` then `java HelloJava` from the folder that contains the class (or with a correct classpath).

Q5. Why is Java popular for hiring in India?

A: Large existing codebases, strong Core Java interviews, banks and product companies, and a clear path into REST backends.